

Algorithms Problem Solving

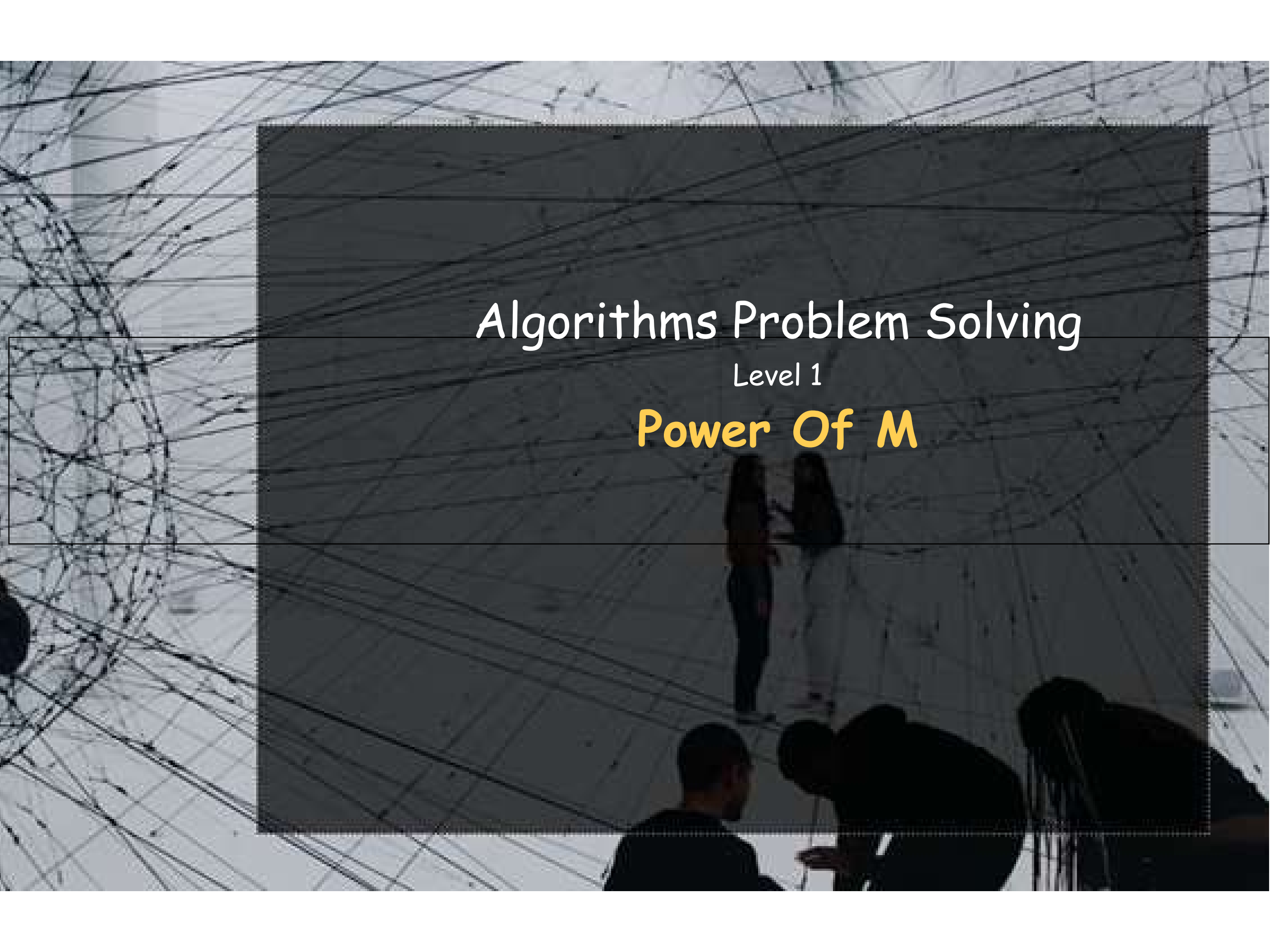
Level 1

Power Of M

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🔍 Definition

- ◆ This program asks the user to enter a base number N and an exponent M , then calculates N raised to the power of M .
- ◆ It calculates the power manually using repeated multiplication instead of using `pow()`.
- ◆ The program uses a `while` loop to perform the multiplication M times.

Explanation

- ◆ `readNumbers()` reads the base `N` and exponent `M`.
- ◆ `power()` starts `result` at 1.
- ◆ While `expo` is not zero, the function multiplies `result` by `base`.
- ◆ `--expo` decreases the exponent by 1 after each multiplication.
- ◆ When `expo` reaches 0, the loop stops and the result is returned.
- ◆ Starting `result` at 1 is important because multiplying by 1 does not change the value.
- ◆ `printResult()` receives the base, exponent, and calculated result and prints them.
- ◆ `printResult(N, M, power(N, M))` calculates the power first and passes the result to `printResult()`.

Steps in main()

1. Read the base N .
2. Read the exponent M .
3. Set `result` to 1.
4. Check whether `expo` is not zero.
5. Multiply `result` by `base`.
6. Decrease `expo` by 1.
7. Repeat until `expo` becomes 0.
8. Return the result.
9. Print the calculated power.

💡 Example

◆ If the user enters:

Base N: 3

Exponent M: 4

◆ The calculation is:

$$1 \times 3 = 3$$

$$3 \times 3 = 9$$

$$9 \times 3 = 27$$

$$27 \times 3 = 81$$

➡ Result

3 to the power of 4 is 81

Note

- ◆ The current implementation does not handle negative exponents correctly.
- ◆ If ``M`` is negative, ``expo != 0`` remains true and ``--expo`` makes it even more negative, so the loop never ends.
- ◆ Negative exponents would also require a fractional result in general.

⊘ Warning

- ◆ If N is less than 1, the loop does not execute and the result remains 0.



Comparison

▨ ``power()`` with ``while``

- ✓ Calculates the power using repeated multiplication.
- ✓ Does not require ``pow()``.
- ✓ Clearly demonstrates how exponentiation works.

▤ ``pow()``

- ✓ Provided by the ``<cmath>`` library.
- ✓ Uses ``pow(base, exponent)``.
- ✓ Can directly calculate the power without writing the multiplication loop.

✦ Summary

- ✓ The program calculates N to the power of M .
- ✓ ``result`` starts at 1.
- ✓ The ``while`` loop repeatedly multiplies the result by the base.
- ✓ ``--expo`` decreases the number of remaining multiplications.
- ✓ ``M = 0`` correctly produces 1.
- ✓ The current code should not be used with negative exponents because they are not handled.

Thank you 😊